

FIRST AID



- The initial process of assessing and addressing the needs of someone who is experiencing medical emergencies.
- Allows a “non-medical expert” to quickly determine a person’s physical condition and the course of treatment.
- Can make a difference to a person’s recovery and could save their life.

Definition of first aid:

- First aid is the provision of **immediate care** to a victim with an injury or illness, usually effected by a **lay person**, and performed within a **limited skill range**.
- First aid is normally performed until the injury or illness is **satisfactorily dealt** with (such as in the case of small cuts, minor bruises, and blisters) or until the next level of care, such as a paramedic or doctor, arrives
- First aid is an emergency aid or treatment given to someone injured, suddenly ill, etc., before **regular medical services** arrive or can be reached.

Objectives of first aid/guiding principle:

- The key guiding principles and purpose of first aid, is often given in the mnemonic "3 Ps". These three points govern all the actions undertaken by a first aider.
- **Prevent** further injury
- **Preserve** life
- **Promote** recovery

Principles of first aid

- **Preserve life:**
 - Airway
 - Breathing
 - Circulation
- **Prevent deterioration:**
 - Stop bleeding
 - Treat shock
 - Treat other injuries
- **Promote recovery:**
 - Reassure
 - Relieve pain
 - Handle with care
 - Protect from harm

Philosophy of First Aid

- In the pre-hospital setting, the key contributors to survival and recovery from illness and injury are prompt and effective maintenance of the **body's primary functions**:
 1. Airway
 2. Breathing
 3. Circulation
 4. Bleeding control (life threatening)

Responsibility of a First Aider

- *To assess the situation quickly and safely and call for appropriate help.*
- *To identify the level of injury or the nature of illness affecting the casualty / victim.*
- *To give early and appropriate treatment in a sensible order of priority.*
- *To make and pass on a report, give a further help if its required.*

NOSE BLEED



The Do and Don'ts for First Aid Treatment

DIAGNOSIS



Nosebleed



- Do not lean back.
- Leaning back can be harmful as the blood could block the windpipe, blocking the airway.
- Sit in a comfortable upright position and lean forward slightly.
- Then pinch your nose just below the bony nose bridge and above the fleshy lobes of the nostrils until the bleeding is stemmed.
- Aftercare: Once the bleeding is controlled, do not blow your nose as this might dislodge the clot and make you bleed again.

CHOKING

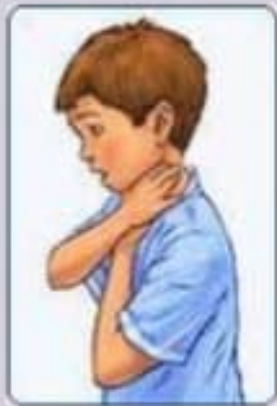


The Do and Don'ts for First Aid Treatment

DIAGNOSIS



CHOKING



ASK! Are you choking?

If the victim able to talk, groan, wheeze or cough, he is partially choked.

- Slap the victim back's hard.
- Using your fingers to force out the item out of the victim's mouth.



- Remain calm and encourage the victim to keep coughing to try and clear the blockage.
- Stand slightly behind the person to one side.
- Support their chest with one hand. Lean the person forward so that the object blocking the airway will come out of their mouth, rather than going further down.
- Give at least 5 sharp blows between the person's shoulder blades with the heel on your hand.

Stop after each blows to check if the blockage has cleared. If not, give up to five abdominal thrusts.

CHOKING



The Do and Don'ts for First Aid Treatment

DIAGNOSIS



CHOKING



Complete blockage

If the victim unable to make any sound at all.

- Using your fingers to force out the item out of the victim's mouth.

Steps in Abdominal thrusts

1. Stand behind the person who is choking.
2. Place your arms around the waist and bend them well forward.
3. Clench your fist and place it right and place it right above the person's navel (belly button)
4. Place your other hand on top, thrust both hands backwards into their stomach with a hard, upward movement.

Do it five times (1 cycle), stop each cycle to check if the blockage has been cleared.

Alert!

- Do not thrusts on pregnant and on a very large sized adult.

BURNS



Burns

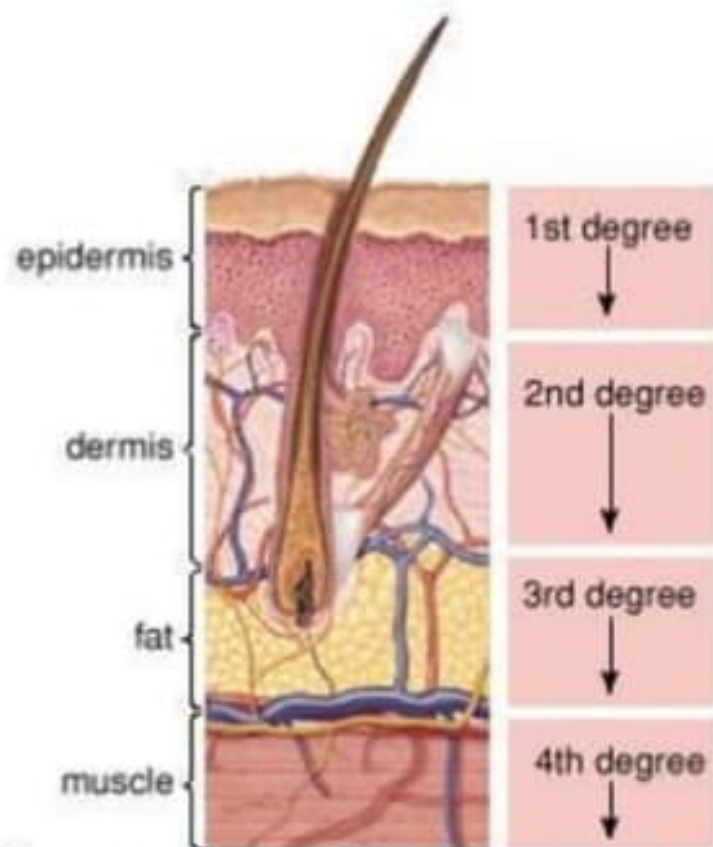
Definition: A burn is damage to the skin or underlying tissue caused by heat. There are 3 levels of severity; 1st (Superficial), 2nd (Partial thickness), 3rd (Full thickness).

Causes: There are 5 main sources of burns;

- Electricity,
- Radiation (sun),
- Thermal (something hot),
- Chemical,
- Friction

Prevention:

- Use safety rules.
- Use safety equipment when working with chemicals.
- Hire professionals for work dealing with e.g. electricity.
- Avoid sun exposure.
- Keep hot objects away from children.



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Management:

- For 1st and 2nd degree burns you should cool the area immediately with gently running cold water for about 10-15 minutes, or until it has cooled off. Do not break any blisters as this will make the wound worse.
- For 3rd degree burns do not put anything on the burn, seek medical help immediately and treat for shock. 3rd degree is extremely life threatening even when a small body part is affected. If there is clothing on the burn do not remove it as this may also remove skin. There is a very high risk of infection from this kind of burn.

First Aid for BURNS

1 Cool with running water

Cool burned area with cool-running water for up to 30 minutes, or until pain is less severe. Take off any jewellery and clothing that could be in the way. Do not attempt to peel any clothing that is stuck to the wound as this could cause the skin to tear. If running water is not available, use a damp cloth, wet towel, sponge or immersion in water. May shock, aggravate or increase pain. May shock, aggravate or increase pain.

Minor burns with only redness and no blisters, can be treated with first aid burn treatment or spray.



Do not use ice or ice water, which can cause tissue damage. Do not apply butter, oil, toothpaste, cream, ointment or a burn dressing unless it is specifically recommended by the manufacturer.



2 Cover the burn

Cover the burn with sterile material to protect from infection. Use a clean, dry dressing or plastic cling film wrap to cover the burn.



Do not use adhesive or fluffy dressings. Do not break blisters or remove partial skin.



3 Call emergency assistant

For deep or extensive burns of any sort, send the patient immediately to the hospital for further medical treatment.



Notes:

- As with all other emergencies make sure the area is safe for you first. Watch out for live wires, hot objects, chemical spills, etc.
- The severity of a burn can also be increased depending on; Which part of the body is affected, e.g. face, and neck.
- The amount of the body that is burnt, e.g. only fingertip or entire arm.
- With electrical burns check for an exit wound as well as treating for the entrance wound.
- With chemical burns flush the area with lots of water to get it off the casualty's skin.
- Never apply ointments, butter, or other home remedies on burns, as this may make the burn worse, keep the heat trapped in, or cause an infection.

Wound

Seek medical care for a cut or wound if:

- It's jagged or deep, or if it is a puncture wound.
- it's on the face
- it's the result of an animal bite
- there is dirt that won't come out after washing
- the bleeding will not stop

Wounds

- Open Wounds
 - A break in the skin's surface that results in external bleeding and may allow bacteria to enter the body that can cause infection
 - **Abrasion**
 - The top layer of skin is removed with little or no blood loss
 - Scrape
 - **Laceration**
 - A cut skin with jagged, irregular edges and caused by a forceful tearing away of skin tissue
 - **Incisions**
 - Smooth edges and resemble a surgical or paper cut



Wounds

- **Punctures**

- Deep, narrow wounds such as a stab wound from a nail or a knife in the skin and underlying organs

- **Avulsion**

- Flap of skin is torn loose and is either hanging from the body or completely removed

- **Amputation**

- Cutting or tearing off of a body part such as a finger, toe, hand, foot, arm, or leg

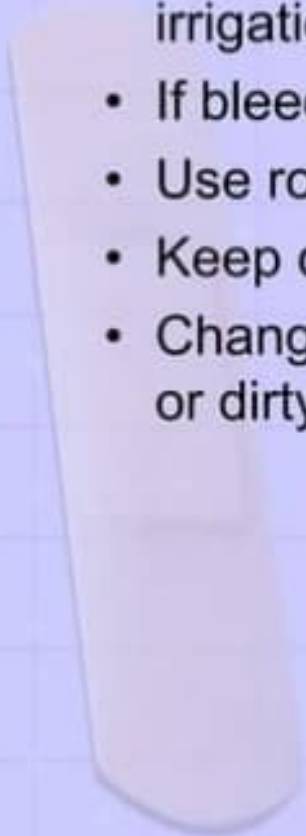


Wounds

- What to Do:
 - Wear gloves (if possible) and expose wound
 - Control bleeding
 - Clean wounds
 - To prevent infection
 - Wash shallow wound gently with soap and water
 - Wash from the center out / Irrigate with water
 - Severe wound?
 - Clean only after bleeding has stopped

Wounds

- Wounds Care
 - Remove small objects that do not flush out by irrigation with sterile tweezers.
 - If bleeding restarts, apply direct pressure.
 - Use roller bandages (or tape dressing to the body)
 - Keep dressings dry and clean
 - Change the dressing daily, or more often if it gets wet or dirty.



Wounds

- Signs of Wound Infection:
 - Swelling, and redness around the wound
 - A sensation of warmth
 - Throbbing pain
 - **Fever / chills**
 - **Swollen lymph nodes**
 - **Red streaks**
 - **Tetanus (lock jaw), should receive injection in first 72 hours.**

Dressings and Bandages

- The purpose of a dressing is to:
 - Control bleeding
 - Prevent infection and contamination
 - Absorb blood and fluid drainage
 - Protect the wound from further injury
- What to Do:
 - Always wear gloves (if possible)
 - Use a dressing large enough to extend beyond the wound's edges.
 - Cover the dressing with bandages.

Broken bones and fractures

- **Definition:** A break or crack in a bone is called a fracture.
- In most cases the damage to the bone will be under the skin, which is called a closed fracture,
- But sometimes bits of the bone can puncture through the skin to become an open fracture.
- In both cases you'll need to treat the casualty for [shock](#).
- Even if you can't see any blood, the break might have caused some internal bleeding.
- To break a fully grown bone, a huge amount of force is needed
- But bones that are still growing are supple and can split, crack or bend quite easily, a bit like a twig.

Fracture

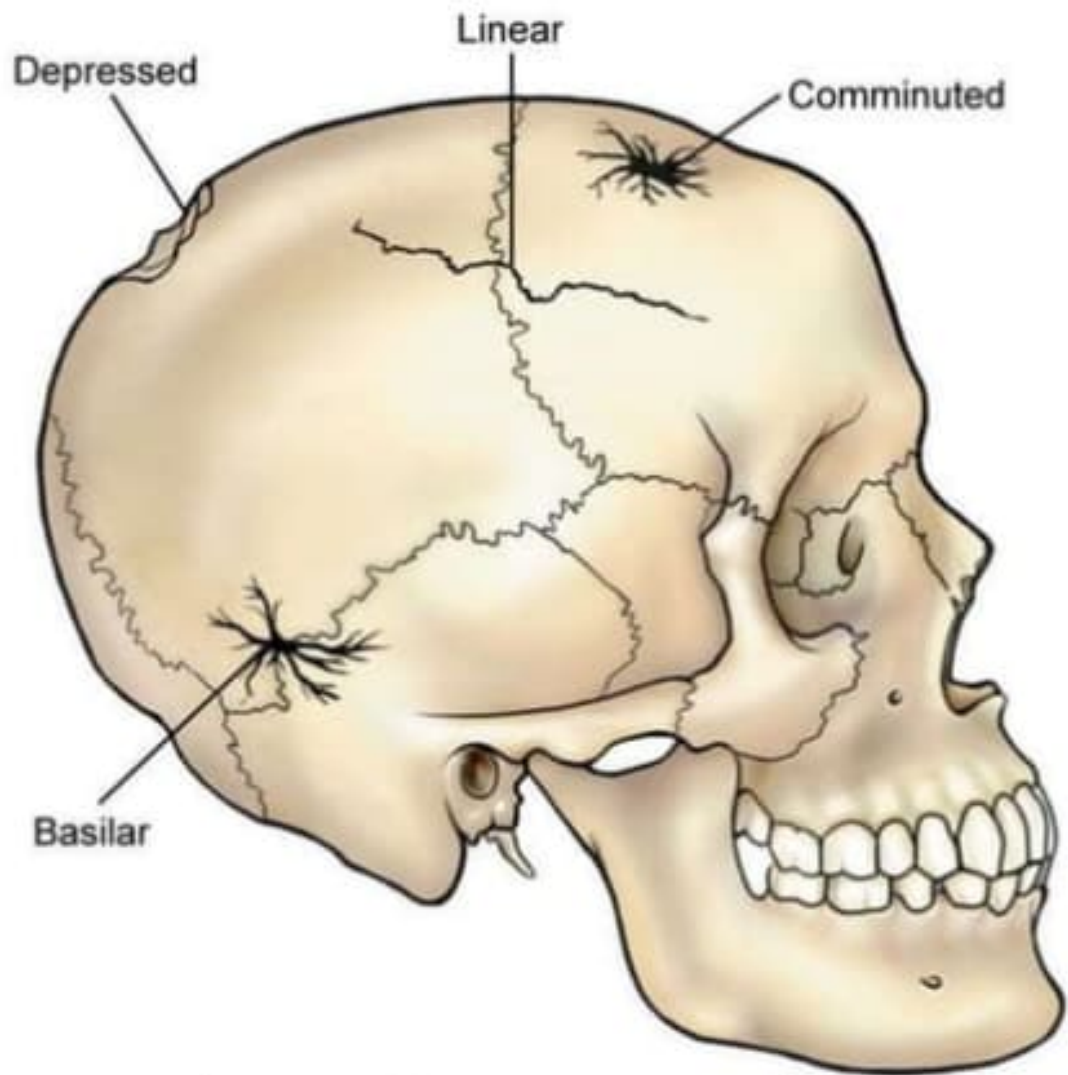
Signs:

- The seven things to look for are:
- Swelling
- Difficulty moving
- Movement in an unnatural direction
- A limb that looks shorter, twisted or bent
- A grating noise or feeling
- Loss of strength
- Shock

Types of fracture:

- **Green stick fractures:** Closed fracture mostly it occurs in children..
- **Complicated fractures:** They occur when the jagged ends of the bone fragments damage blood vessels, nerves or a joint, broken bones in the chest may penetrate the lung, heart or liver. In fractures of the skull the brain is usually damaged.
- **Depressed fractures:** These occur in the skull when the broken ends of the bones are pressed inwards.
- **Comminuted Fractures:** In these cases, the bone is broken into several fragments. This is serious because there will be muscle damage with more bleeding at the fracture site.

Types of skull fractures



[From Mosher, F., & Neighbors, M. (1998). *Medical-surgical nursing: Foundations for clinical practice* (2nd ed.). Philadelphia: Saunders.]
Fig. 56-26. Skull fractures.

Types of fracture:

- **Impacted Fractures:** After a heavy fall, the fracture may be impacted by the force, (eg.) Spinal injury falling from tree.
- **Pathological Fractures:** These occur when the bone is weakened by loss of calcium, infection or cancer. Minimal cause a break in such cases. In old age the bones are more brittle, and may break spontaneously due to calcium loss which is part of the ageing process.
- **Stress Fractures:** Stress caused by repeated minor trauma as in athletic training. Involved in strenuous training, such as jogging or marathon running.

Management:

- If it is an open fracture, **cover** the **wound** with a sterile dressing and secure it with a **bandage**. **Apply pressure** around the wound to control any bleeding.
- **Support** the injured body part to stop it from moving. This should ease any pain and **prevent** any further **damage**.
- Once you've done this, call medical help. While waiting for help to arrive, **don't move** them unless they're in immediate danger.
- **Protect** the injured area by using **bandages** to secure it to an **uninjured part** of the body to stop it from moving. For example, fractures on the arm can be secured with a sling, and a leg with a fracture can be tied to the uninjured leg.
- Keep checking the casualty for signs of **shock**. This does not mean emotional shock, but is a life-threatening condition, often caused by losing blood.
- If they lose responsiveness at any point, open their airway, check their breathing and prepare to treat someone who's become **unresponsive**.

Splints:

- The main purpose of a splint is to keep an injured body part immobilized (e.g. a broken leg).
- It should only be done if paramedics are going to take a long time arriving, or if you have to move the person.
- Never move or try to realign the injured body part. Always splint in the position found.

There are three main types of splints:

- 1. **Anatomical**: this means using a non-injured body part to immobilize an injured body part. E.g. to splint a broken left lower leg you would tie both legs together so the good leg provides support to the injured leg.
- 2. **Soft**: this means using something like a thick sweat shirt, a jacket, a towel, or a blanket to wrap around and immobilize. This kind of splint works very well with hand or ankle/foot injuries.
- 3. **Rigid**: this refers to using a firm object, such as a piece of wood, to immobilize. There are many types of rigid splints you can purchase or you can use whatever you see around you e.g. magazines, newspapers, umbrellas, etc.

Injuries to the soft tissues:

- **Definition:**

- This is an injury to a bone, a joint, a ligament, or a tendon.
- Joint injuries usually involve a dislocation.
- This is where the bone has popped out of its socket.
- This may be accompanied with a fracture, a strain, or a sprain.
- It may pop back in it may not.
- Either way seek medical help.
- Do not push it back into place.

Injuries to the soft tissues:

- Tendons are strong tissues that connect a muscle onto a bone.
- When a tendon tears it is called a strain.
- When they become torn they take a very long time to heal, many times never as good as before, and sometimes surgery is required to reattach them.
- When a ligament is torn it is called a sprain.
- Ligaments connect a bone to another bone.
- These are found around the joints.
- Ligaments are very strong, but, as with tendons, when they tear they take a long time to heal, never as good as before, and sometimes surgery is required.

Injuries to the soft tissues:

- **Causes:**

- Any kind of force that is greater than what the tissue can withhold will cause such an injury. Some common activities include falling, twisting, getting hit, etc.

- **Prevention:**

- Use safety equipment and wear it properly.
- Use seat belts and car seats.
- Keep joints and bones strong through weight bearing physical activities.

Injuries to the soft tissues:

Signs/Symptoms:

- A 'snapping' noise.
- Pain.
- Deformity.
- Inability to move.
- Swelling.
- Bruising.

Injuries to the soft tissues::

Management:

- Apply the **RICE** principle.
- **Rest** the injured body part and the entire casualty.
- **Immobilize** the injured body part.
- **Cold compress** over the injury to reduce swelling.
- **Elevate** the injured body part if it can be done without causing further injury.
 - Seek medical help.
 - Do not rub or move the injured body part.
 - If there is a protruding bone then bleeding will need to be taken care of by applying indirect pressure.
 - Never straighten or realign an injured body part.
- **Notes:** Bone fractures, if set properly, will heal fully in a few weeks and will be stronger than before. This happens because the area builds up with more calcification than before.



Cramp



1. **Cramps** : Painful muscular involuntary contraction

A muscle **cramp** is a strong, painful contraction or tightening of a muscle that comes on suddenly and lasts from a few seconds to several minutes. It often occurs in the legs. A muscle **cramp** is also called a charley horse. Nighttime leg **cramps** are usually sudden spasms, or tightening, of muscles in the calf.

s/s:

- . -Sweating or moist skin. -Tired, irritable, and thirsty.



Causes of Leg Cramps:

- Dehydration or inadequate intake of water.
- Depleted levels of potassium and sodium. (salt)
- Depleted carbohydrate levels.
- Tense or stiff muscles.
- Vitamin deficiencies may also cause cramps.
- Poor blood circulation also causes cramps.

Treatment:

Removing them from the heat. -Gently stretch/massage affected area.

- Slowly rehydrate with water, juice, or sport drinks. Rest for a couple of hours.
- Avoid alcohol, caffeinated and/or carbonated drinks.

Cardiopulmonary resuscitation:

- **Airway:**
- Place them on their back, carefully so as to not cause any injury.
- Open their airway by tilting their head back and lifting their chin upwards. This will remove the tongue from blocking the airway.
- Keep the airway open.
- **Breathing:**
- Check for Breathing by looking, listening, and feeling for air (10 seconds).
- If they are breathing then monitor and put them in the recovery position until the paramedics arrive.
- If they are not breathing give them 2 breaths, allowing the air to come out in between.
- If the air goes in then go to "Circulation."
- If the air does not go in, re-position the head-tilt/chin-lift a bit further back and try blowing again. Be careful not to injure their neck.
- If the air still does not go in then go to "Circulation" but check the mouth for the food after doing CPR.

HOW TO OPEN AN AIRWAY

- Tilt Head and Back and Lift Chin up with fingers under the jaw to establish Airway (Move head as little as possible if there may be a neck injury.)



- **Circulation:**

- Start CPR if needed (compressions and breaths).
- It is 30 compressions to 2 breaths. Continue until paramedics arrive or until something changes with the person (reassess at this point).
- If the air was not going in make sure you check the mouth, to see if the food came out, after each set of compressions. If you see the object in their mouth take it out and reassess breathing.
- Under this category, we are also concerned with shock and severe bleeding.

- **About compressions:**
 - Adult: use both hands. Compress 4-5 cm deep.
 - Child: use one hand. Compress almost half way down.
 - Infant: use two fingers. Compress almost halfway down.
 - Fast and smooth.
 - Do not stop unless something changes.
 - Ribs may break, keep going.
 - If they vomit roll them onto their side, clean their mouth, continue.
- **Two-Rescuer CPR:** In a situation where there are 2 trained first aiders, and they work well together, one rescuer can perform the compressions while the other rescuer gives the breaths. All the steps are the same, nothing changes. This is a bit more sufficient and less tiring for the rescuers. If the rescuer doing the compressions becomes tired they can switch positions.

Bites and stings:

Signs/Symptoms: Wound (animal or human bite) or swelling and pain (insect sting).

Treatment:

- For animal and human bites, cleanse wound with soap and water and apply iodine containing antiseptic; submit animal for rabies test.
- Prevent exertion and taking of stimulants by victim. For insect stings apply cortisone ointments, soothing lotions, or cool compress.
- Persons who are allergic to insect stings should carry adrenaline with them at all times.



Snake bite: Not all snakes are poisonous, but if you get bitten by a poisonous one you better rush to medical help immediately. Because snake anti-venom depends on the type of snake that caused the bite, if you can, try and remember the color and pattern of the snake so they can identify it and give you the correct anti-venom.

Treatment:

- Have the person **rest** with the bitten body part lower than the heart.
- For poisonous snakebite, cooling the site of the wound with **ice** will slow down absorption of poisons; antivenin treatment is required only for a small number of reptile bites.
- Seek medical help as soon as possible.
- If the person becomes unconscious begin the Primary Survey.

Prevention:

- If you will be in an area where there may be snakes;
- Wear **high boots** that cover your calves.
- Make noise with your feet as you walk to **scare off** any snakes.
- If there are trees look on the branches as you walk.
- Check ahead of time what types of snakes might be in the area. So if a bite does occur you can notify the hospital what kind of snake it was.
- Check ahead of time with local hospitals to see if they do have snake anti-venom.

Notes:

- Unless you want poison in your mouth do NOT cut the bite site and suck the blood out.
- Do not try to catch the snake it will bite again.
- There are many snake bite kits on the market, all designed to suck the poison out.
- This may work somewhat, but, blood flows very quickly and the poison may spread very quickly.
- Do NOT tie a tourniquet to try and stop the blood from flowing. This may cause death to the limb from lack of blood flow.



Dog Bite: A rabid animal is aggressive, drooling saliva, must be approached with care. The most serious infection risk is rabies, a potentially fatal viral infection of the nervous system. The virus is carried in the saliva of the infected animal. Tetanus is also a potential risk following any animal bite. The dog should be watched for 10 days. If the dog is healthy after this period then there is no danger of rabies.

Symptoms:

- Headache, nausea, vomiting, fever.
- Restlessness, confusion.
- Difficulty in swallowing.
- Foul smelling of the mouth.
- Hydrophobia
- Difficulty in drinking water.
- Aerophobia
- Respiratory paralysis.

Management:

- Thorough washing of the bitten area and wash the wound with soap and water (Detergent solution) for 5-10minutes.
- Dress the wound with clean sterile gauze or cloth. Apply antibiotic cream
- Shift the causality immediately to the hospital
- The dog should be kept under observation for 10 days

Cat bite: Cats keep roaming around the houses but if they are disturbed and touched they become VIOLENT and can attack. It causes two types of wounds

- 1. On biting with teeth 2. Scratches by the nails
- Management:
- Wash the wound and apply the dressing properly.

General Principles

- Reassure the casualty and explain to him what you are planning to do
- Seek for assistance
- No more than one leader should be there to guide the team.
- Do not try to move a seriously injured victim alone when assistance is available.

Cont..

- Maintain proper body mechanics while moving the victim.
- Take immediate action to transport the casualty to the nearest health care agency.
- Before transporting consider the nature and severity of injury, first aid already provided, number of assistants available and the distance and route to be covered



Methods of transport

- 
- When only one first aider is available and the casualty is with minor injuries.

1. Cradle method

- Carrying or lifting the victim in hands



Fig: 15.36 ■ Cradle Method

2. Human crutch method

- First aider himself acts as a support to the casualty
- He stands close to the victim and holds his waist.
- He then keeps the casualty's hand on his neck and lift him with the other hand and gives him support
- Used only when the victim is conscious and co-operative with only minor wounds.



Fig: 15.37 ■ Human Crutch

3. Fireman's lift and carry method

- A)

With the injured person standing, the first aider holds one of his hands and with the other hand holds his legs over the knee and carry him on the back.

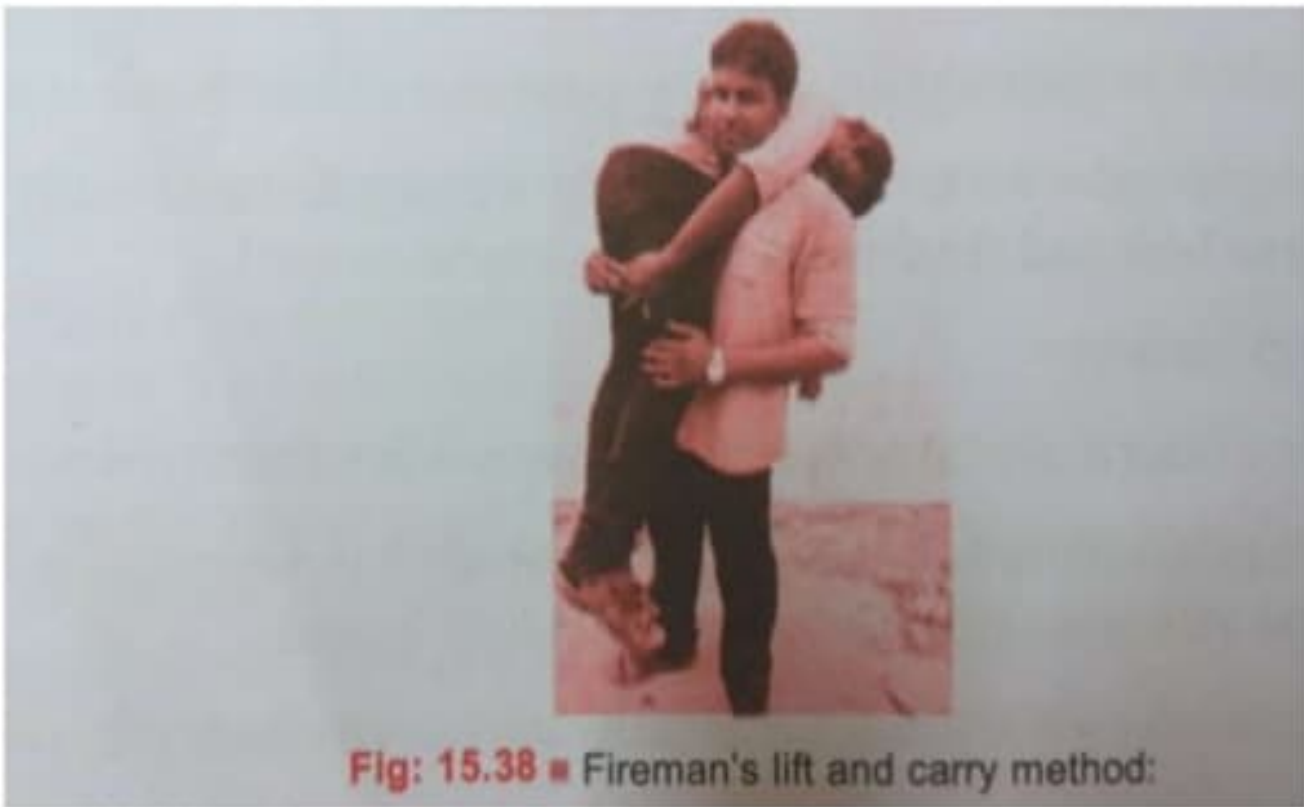


Fig: 15.38 ■ Fireman's lift and carry method:

- B) Pick a back method

Help the casualty to sit on a chair or on a high place. Bend before him and tell him to fold his legs around your waist and hands around the shoulders. Slowly lift him and rise from the sitting position, maintaining body mechanics.



15.39 ■ Pick a back method

4. Drag method

- It is useful in situations when the casualty is unable to stand and when he is under danger and has to be moved quickly from the site of danger
- Eg: fire, electrical shock

Squat at the head end of the casualty. Keep the victim's hands across his chest. Grasp the armpit and cradle the casualty's head on your forearm. Pull the casualty from the site of danger without lifting.

- 
- When two helpers are available and the casualty is with major danger

Two handed seat

- This method is used when the casualty is unable to hold the first aider, may be due to a fracture of hand.
- The two first aiders stand face to face and hold him with one hand of each one at the chest level.
- Hold the other hands together like a hook with fingers.
- Can keep a towel in the hand.
- Slowly lift him from the middle portion of his hands.
- Both should walk now in the same manner slowly.



Four handed seat

- Both persons should stand behind the casualty facing each other.
- Now catch each other's wrist ie. right wrist with left wrist of other person and vice versa.
- After forming the seat, ask the casualty to sit on it with both hands on the shoulders and neck of each first aider.
- Both should now take slow steps.



Fig: 15.41 a, 1.41 b Four Handed Seat

Chair method

- The casualty is made to sit on a chair and fasten him on the chair with a long piece of cloth.
- One person should stand in front and the other person behind the casualty, facing each other.
- Now both the first aiders together carry him slowly, ensuring the casualty does not lean forward or fall down.

Wheel chair method

- The casualty is made to sit on wheel chair as in the ordinary chair.
- Grasp the handles and slowly move the victim

Carrying or lifting the victim with two persons (Fore and AFT method)

- One first aider squat at the head end of the casualty.
- Place the casualty's arms across his abdomen.
- He then catches the victim by encircling his chest while the other person squat between the legs of casualty facing the feet and catches him by putting the hands under the knees.
- Both persons rise together and move slowly.



Fig: 15.42 ■ Fore and AFT method

Sheet/ blanket lift

- The sheet on which the victim is lying can be used, if it is strong enough, for lifting him.

Suspected spinal injury

- Log roll technique

Stretchers

- Devices used to carry/ transport injured patients to a vehicle or to a shelter.

Types

- Standard Furley stretcher
- Pole and canvas stretcher
- Trolley cot

Standard Furley stretcher

- Consists of a canvas/ plastic sheet attached to 2 carrying poles
- There is a stand underneath



Pole and Canvas stretcher

- Consists of a canvas/ plastic sheet about 200cm long and 50cm wide with handles and slide sleeves and a pair of carrying poles
- The poles are passed through the sleeves down the side of the canvas to form the stretcher



Fig. 3.43: A Pole and canvas stretcher

Trolley cot

- It is an adjustable stretcher bed which can be moved on wheels
- Consists of safety means such as side rails, safety straps and wheel brakes



Fig. 8.44: Trolley cot

Adjustable stretcher with wheels



Fig. 8.45: Adjustable Stretcher with wheels

Ambulance stretcher



Fig. 8.46: Ambulance Stretcher